Advanced Representation Learning
(고급표현학습)

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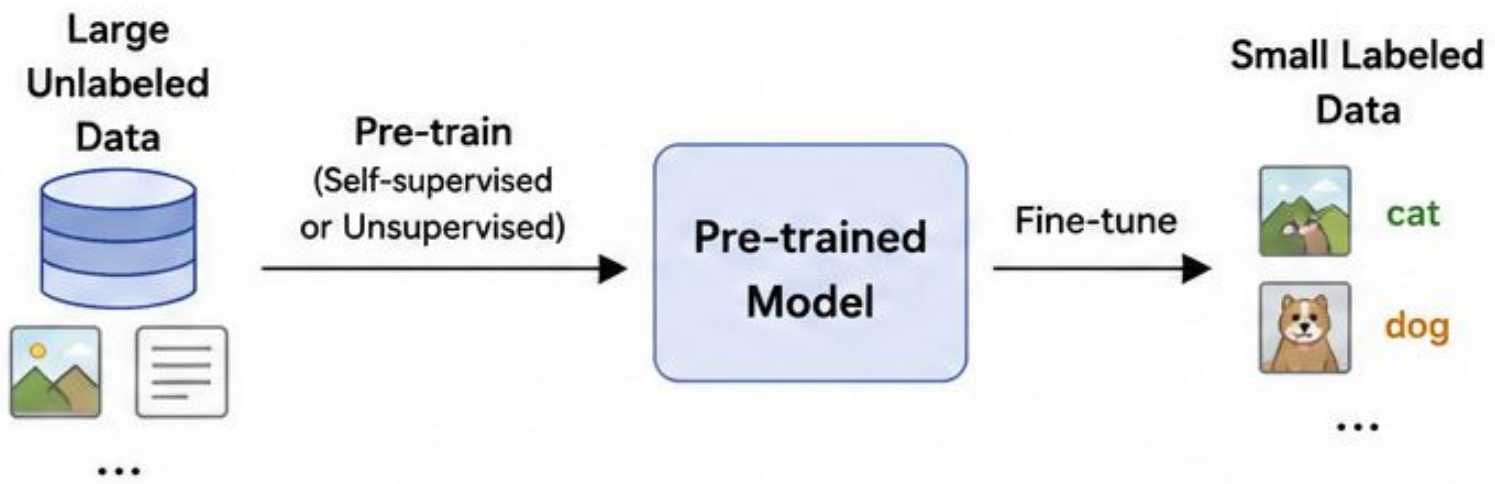
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Semi-Supervised Learning

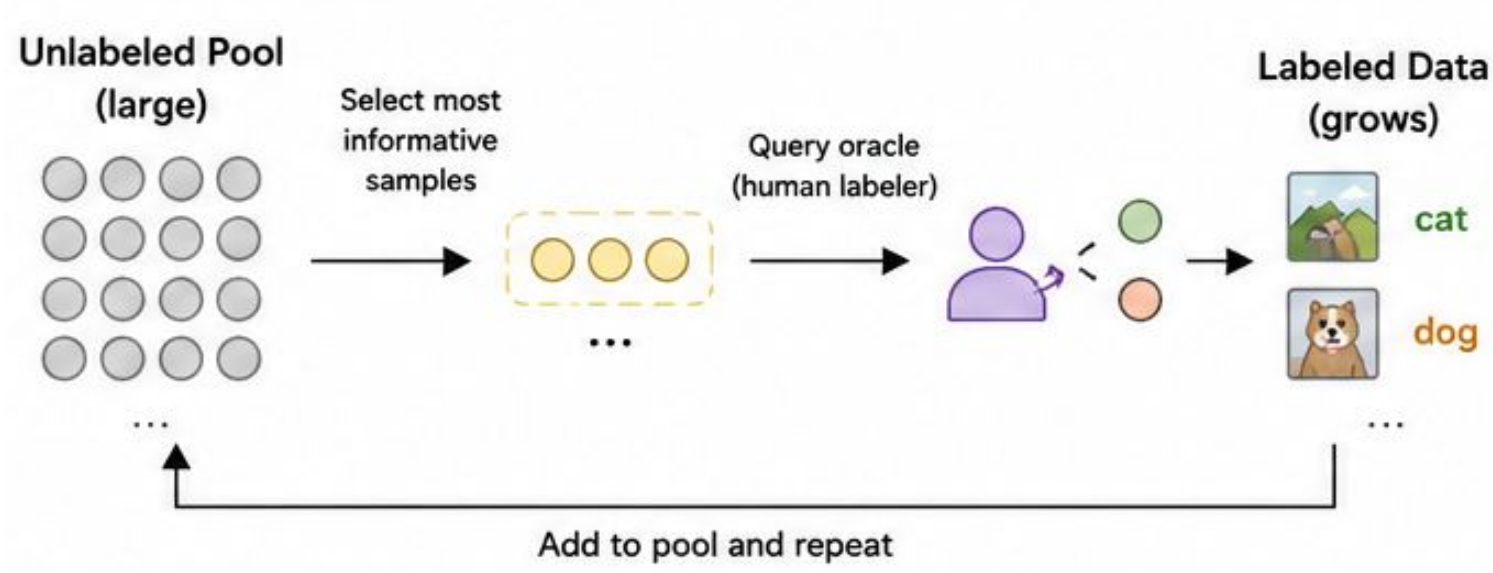
Overcome Limited Amount of Labeled Data

- Pre-training + fine-tuning
  - Pre-train a powerful task-agnostic model on a large unsupervised data corpus
  - Then fine-tune it on the downstream task with a small set of labeled samples



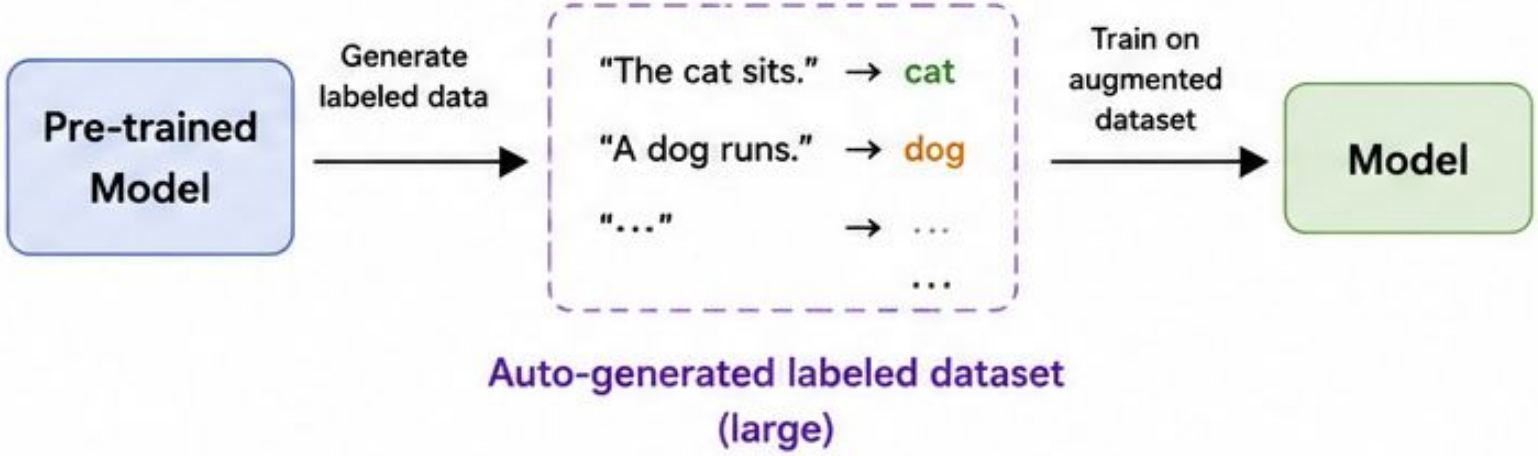
Overcome Limited Amount of Labeled Data

- Active learning
  - Select most valuable unlabeled samples to be collected next
  - Helps us act smartly with a limited budget



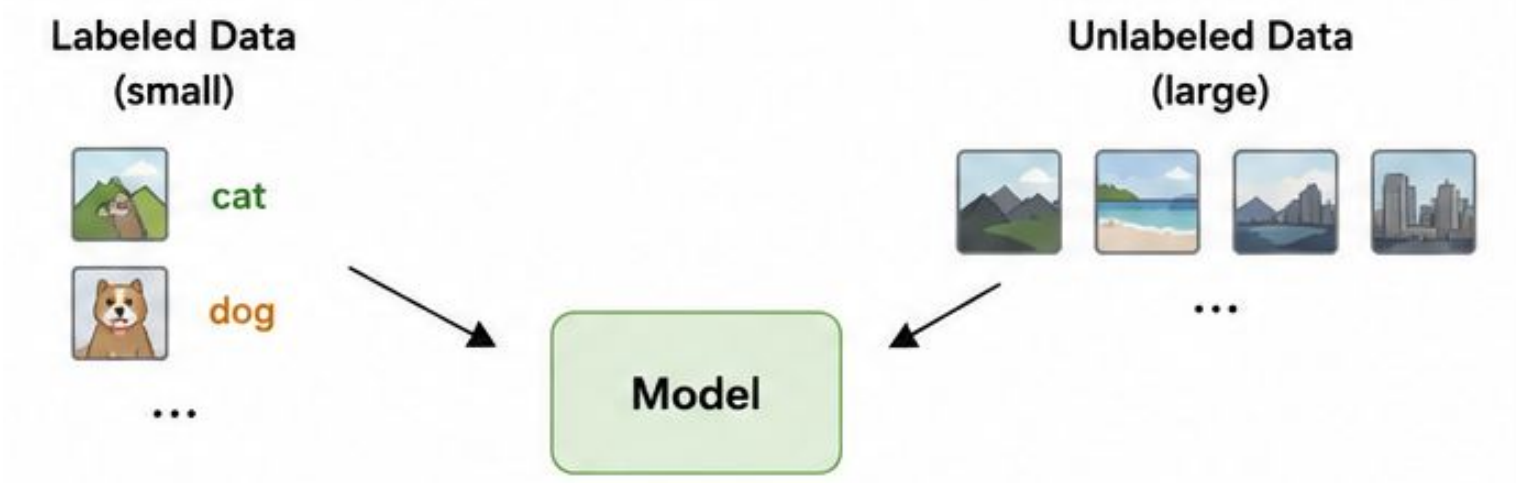
Overcome Limited Amount of Labeled Data

- Pre-training + dataset auto-generation
  - Given a capable pre-trained model, we can utilize it to auto-generate a lot more labeled samples.
  - This has been especially popular within the language domain driven by the success of few-shot learning.



Overcome Limited Amount of Labeled Data

- Semi-supervised learning
  - Learn from the labelled and unlabeled samples together.



Overcome Limited Amount of Labeled Data

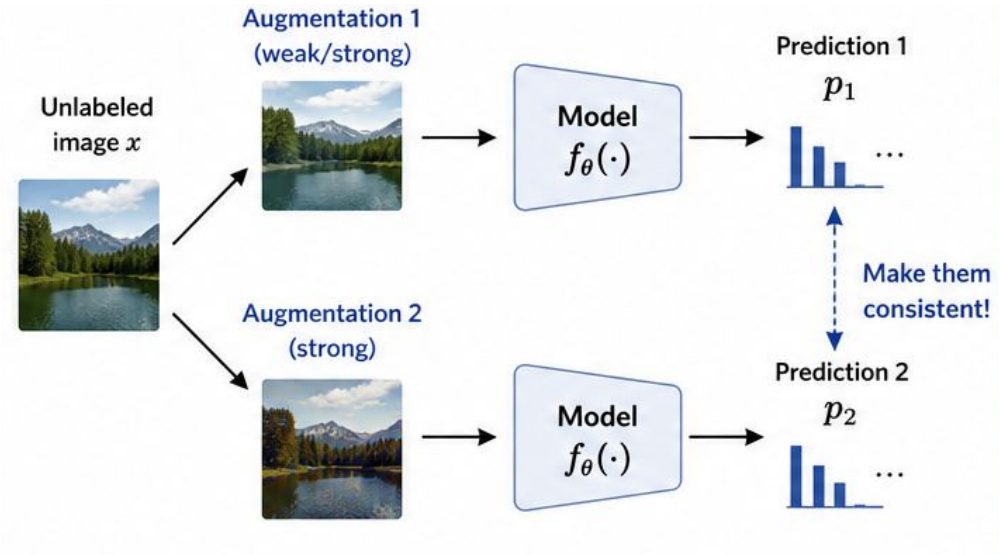
- Semi-supervised Learning
  - Learn from **both labeled and unlabeled data.**
  - Improve generalization when labeled data are scarce.
  - Most SSL methods combine **supervised** and **unsupervised** objectives.

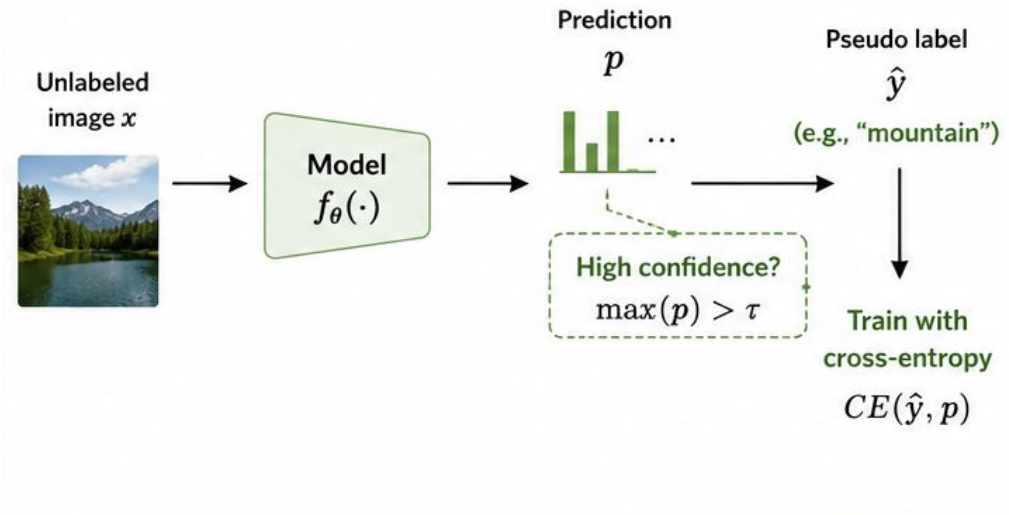
- Learning Objective consists of supervised loss, unsupervised loss, weighting term.

$$\mathcal{L} = \mathcal{L}_s + \mu(t)\mathcal{L}_u$$

Key Question

- How should we design the unsupervised loss?
  - Consistency Regularization
  - Pseudo Label



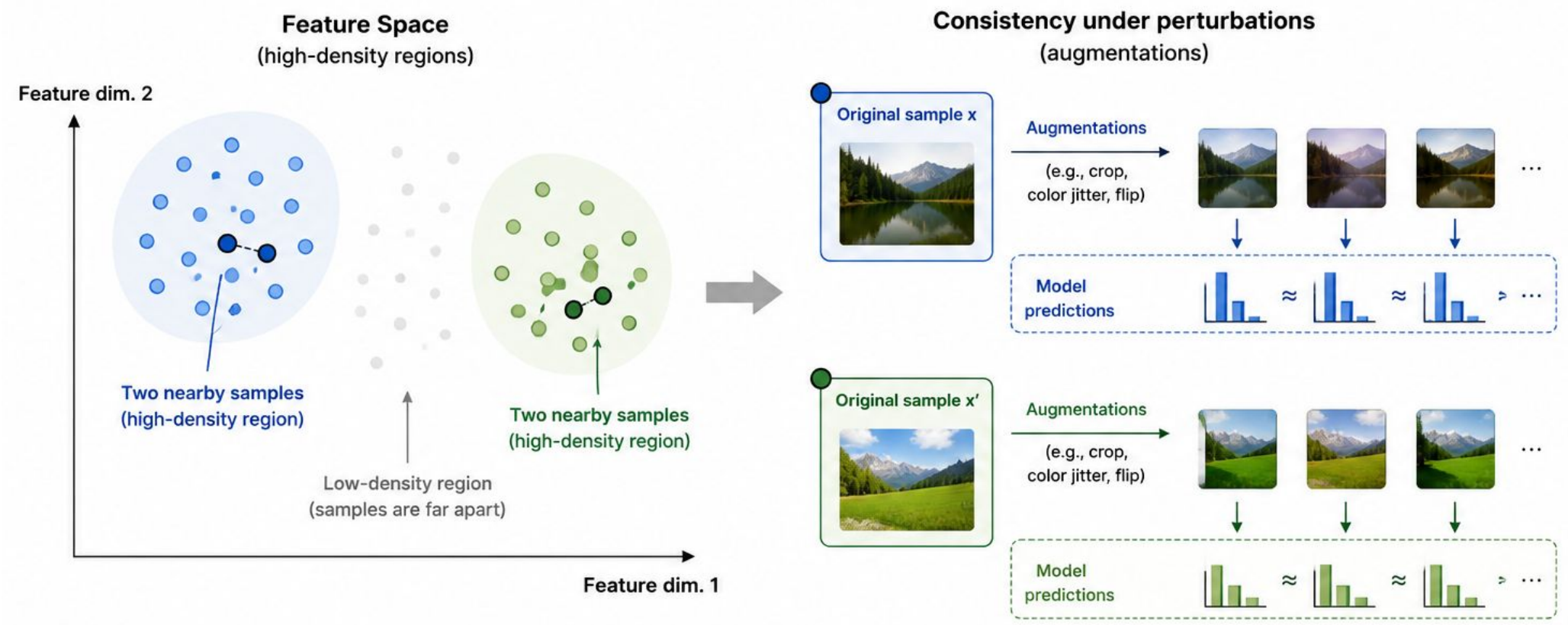


Notations

Symbol	Meaning
L	Number of unique labels.
$(\mathbf{x}^l, y) \sim \mathcal{X}, y \in \{0, 1\}^L$	Labeled dataset. y is a one-hot representation of the true label.
$\mathbf{u} \sim \mathcal{U}$	Unlabeled dataset.
$\mathcal{D} = \mathcal{X} \cup \mathcal{U}$	The entire dataset, including both labeled and unlabeled examples.
$\mathbf{x}$	Any sample which can be either labeled or unlabeled.
$\bar{\mathbf{x}}$	$\mathbf{x}$ with augmentation applied.
$\mathbf{x}_i$	The i-th sample.
$\mathcal{L}, \mathcal{L}_s, \mathcal{L}_u$	Loss, supervised loss, and unsupervised loss.
$\mu(t)$	The unsupervised loss weight, increasing in time.
$p(y\|\mathbf{x}), p_\theta(y\|\mathbf{x})$	The conditional probability over the label set given the input.
$f_\theta(.)$	The implemented neural network with weights θ, the model that we want to train.
$\mathbf{z} = f_\theta(\mathbf{x})$	A vector of logits output by f.
$\hat{y} = \text{softmax}(\mathbf{z})$	The predicted label distribution.
$D[.,.]$	A distance function between two distributions, such as MSE, cross entropy, KL divergence, etc.
β	EMA weighting hyperparameter for teacher model weights.
α, λ	Parameters for MixUp, $\lambda \sim \text{Beta}(\alpha, \alpha)$.
T	Temperature for sharpening the predicted distribution.
τ	A confidence threshold for selecting the qualified prediction.

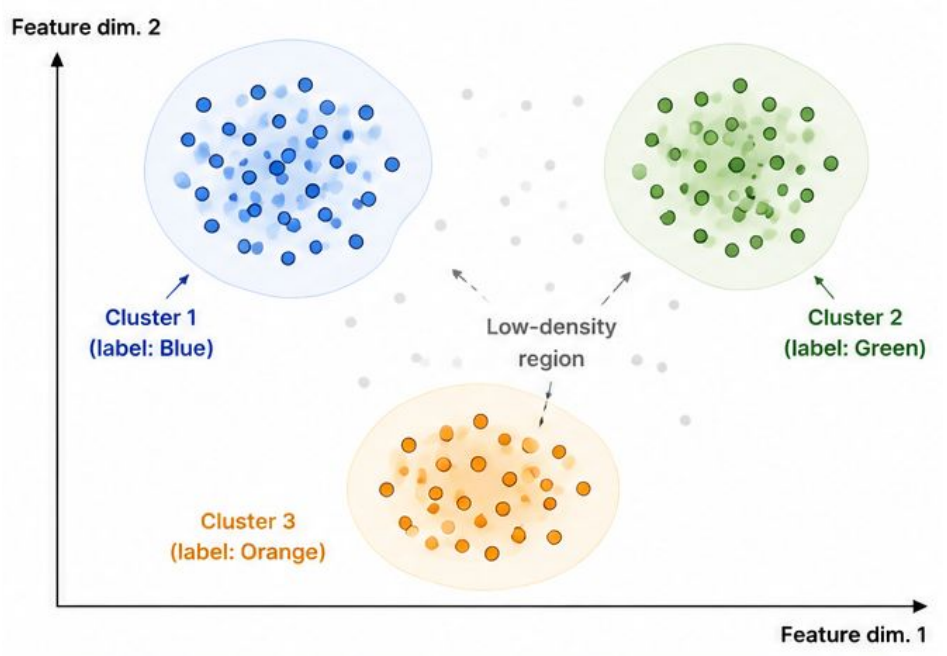
Hypotheses

- H1. Smoothness Assumption
  - Nearby samples should have similar predictions.
  - Small perturbations should not change the label.



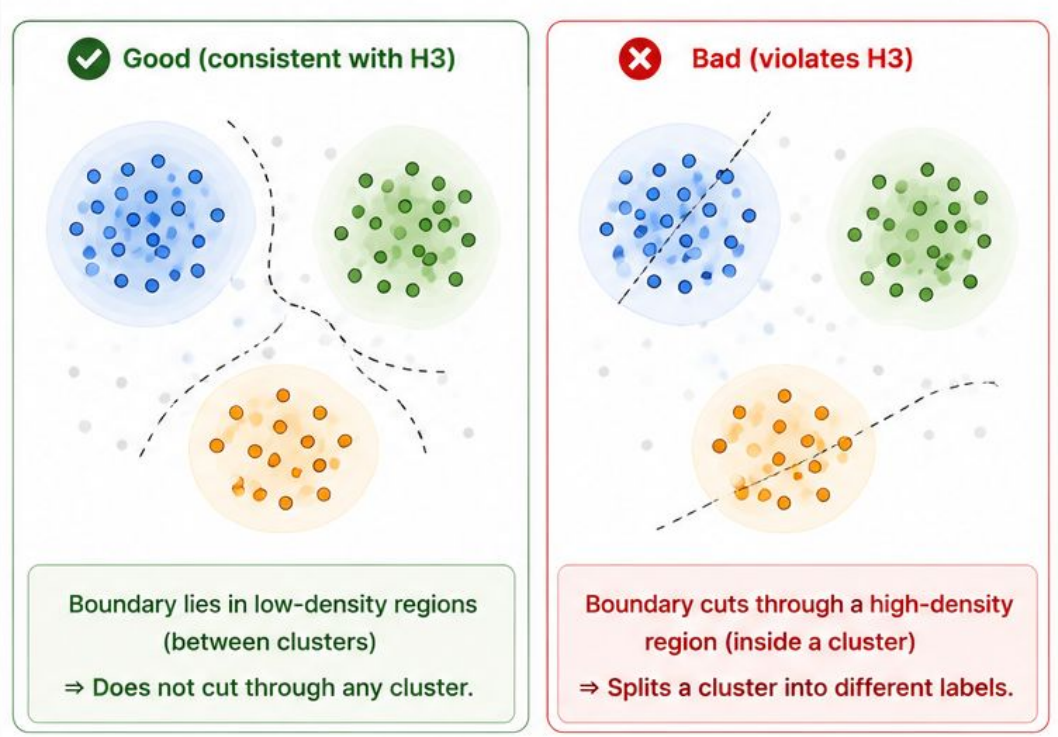
Hypotheses

- H2. Cluster Assumption
  - Samples within the same cluster are likely to share the same label.
  - Decision boundaries should not split a cluster.



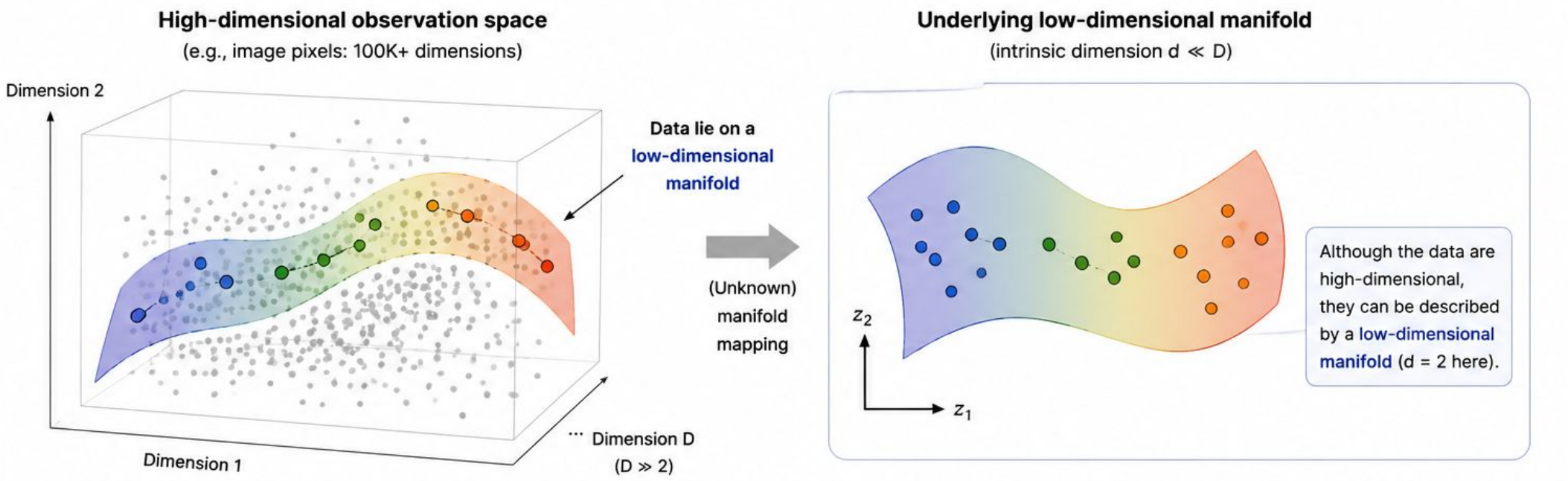
Hypotheses

- H3. Low-Density Separation
  - Decision boundaries should lie in low-density regions.
  - Separate clusters rather than cutting through them.



Hypotheses

- H4. Manifold Assumption
  - High-dimensional data lie on a low-dimensional manifold.
  - Similar samples are connected on the underlying manifold.



Consistency Regularization

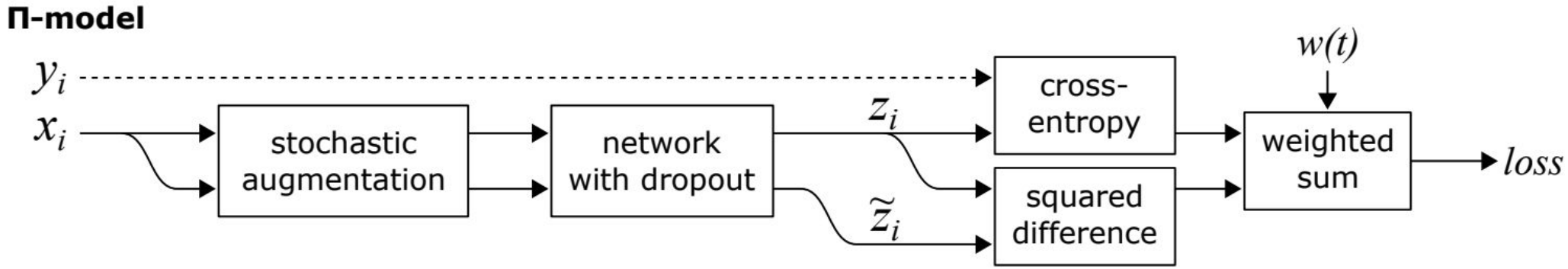
Consistency Regularization

- Core Idea
  - Predictions should remain **consistent under small perturbations.**
  - Perturbations may come from:
    - Data augmentations
    - Model stochasticity (e.g., Dropout)

$$\mathcal{L}_u = D\big(f(x),\ f(\mathcal{A}(x))\big)$$

Π-model

- Two versions of the same input with different stochastic augmentation and dropout masks pass through the network and the outputs are expected to be consistent.



$$\mathcal{L}_u^{\Pi} = \sum_{\mathbf{x} \in \mathcal{D}} \mathrm{MSE}(f_\theta(\mathbf{x}), f'_\theta(\mathbf{x}))$$

Sajjadi et al., ICLR 2016, Regularization With Stochastic Transformations and Perturbations for Deep Semi-Supervised Learning

Π-model

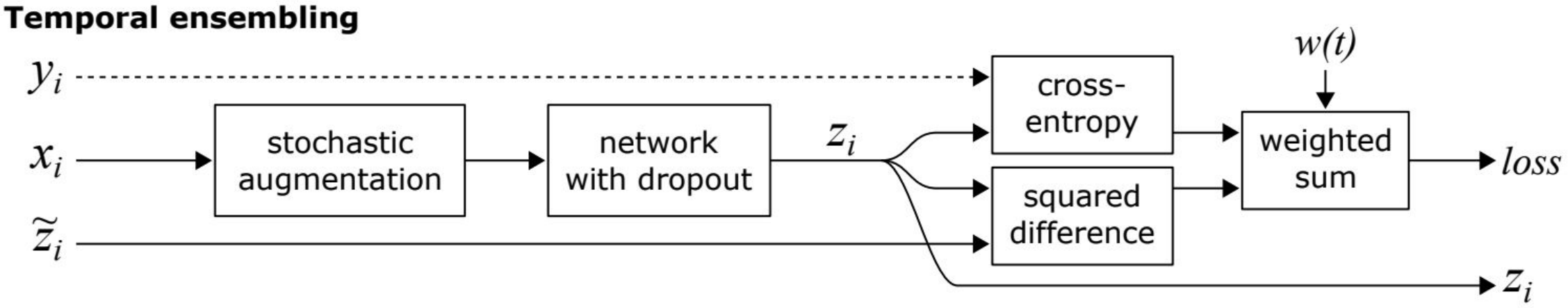
Algorithm 1 Π-model pseudocode.

Require: x_i = training stimuli
Require: L = set of training input indices with known labels
Require: y_i = labels for labeled inputs $i \in L$
Require: $w(t)$ = unsupervised weight ramp-up function
Require: $f_\theta(x)$ = stochastic neural network with trainable parameters θ
Require: $g(x)$ = stochastic input augmentation function
for t in $[1, num_epochs]$ **do**
  for each minibatch B **do**
    $z_{i \in B} \leftarrow f_\theta(g(x_{i \in B}))$ ▷ evaluate network outputs for augmented inputs
    $\tilde{z}_{i \in B} \leftarrow f_\theta(g(x_{i \in B}))$ ▷ again, with different dropout and augmentation
    $loss \leftarrow -\frac{1}{|B|} \sum_{i \in (B \cap L)} \log z_i[y_i]$ ▷ supervised loss component
      $+ w(t) \frac{1}{C|B|} \sum_{i \in B} ||z_i - \tilde{z}_i||^2$ ▷ unsupervised loss component
    update θ using, e.g., ADAM ▷ update network parameters
  end for
end for
return θ

Sajjadi et al., ICLR 2016, Regularization With Stochastic Transformations and Perturbations for Deep Semi-Supervised Learning

Temporal Ensembling

- Π-model requires two forward passes for every input.
- → High computational cost.
- Instead of generating two predictions every iteration,
- → use the historical predictions of the same sample as the consistency target.
- (exponential moving average (EMA) of the model prediction in time per training sample)



Laine et al., ICLR 2017, Temporal Ensembling for Semi-Supervised Learning

Temporal Ensembling

Algorithm 2 Temporal ensembling pseudocode. Note that the updates of Z and $\tilde{z}$ could equally well be done inside the minibatch loop; in this pseudocode they occur between epochs for clarity.

Require: x_i = training stimuli
Require: L = set of training input indices with known labels
Require: y_i = labels for labeled inputs $i \in L$
Require: α = ensembling momentum, $0 \leq \alpha < 1$
Require: $w(t)$ = unsupervised weight ramp-up function
Require: $f_\theta(x)$ = stochastic neural network with trainable parameters θ
Require: $g(x)$ = stochastic input augmentation function

$Z \leftarrow \mathbf{0}_{[N \times C]}$ ▷ initialize ensemble predictions
$\tilde{z} \leftarrow \mathbf{0}_{[N \times C]}$ ▷ initialize target vectors
for t in $[1, num_epochs]$ **do**
  for each minibatch B **do**
    $z_{i \in B} \leftarrow f_\theta(g(x_{i \in B}, t))$ ▷ evaluate network outputs for augmented inputs
    $loss \leftarrow -\frac{1}{|B|} \sum_{i \in (B \cap L)} \log z_i[y_i]$ ▷ supervised loss component
      $+ w(t) \frac{1}{C|B|} \sum_{i \in B} ||z_i - \tilde{z}_i||^2$ ▷ unsupervised loss component
    update θ using, e.g., ADAM ▷ update network parameters
  end for
  $Z \leftarrow \alpha Z + (1 - \alpha) z$ ▷ accumulate ensemble predictions
  $\tilde{z} \leftarrow Z / (1 - \alpha^t)$ ▷ construct target vectors by bias correction
end for
return θ

Laine et al., ICLR 2017, Temporal Ensembling for Semi-Supervised Learning

Temporal Ensembling

Table 1: CIFAR-10 results with 4000 labels, averages of 10 runs (4 runs for all labels).

	Error rate (%) with # labels	
	4000	All (50000)
Supervised-only	35.56 ± 1.59	7.33 ± 0.04
with augmentation	34.85 ± 1.65	6.05 ± 0.15
Conv-Large, Γ-model (Rasmus et al., 2015)	20.40 ± 0.47	
CatGAN (Springenberg, 2016)	19.58 ± 0.58	
GAN of Salimans et al. (2016)	18.63 ± 2.32	
Π-model	16.55 ± 0.29	6.90 ± 0.07
Π-model with augmentation	12.36 ± 0.31	$\mathbf{5.56 \pm 0.10}$
Temporal ensembling with augmentation	$\mathbf{12.16 \pm 0.24}$	5.60 ± 0.10

Laine et al., ICLR 2017, Temporal Ensembling for Semi-Supervised Learning

Temporal Ensembling

- Advantages
  - Only **one forward pass** per iteration
  - More stable target than Π-model
  - Lower computational cost
- Limitation
  - The EMA target is maintained **per training sample.**
  - Requires storing predictions for every sample
  - Updated only **once per epoch**
  - Difficult to scale to large datasets
- Next: Mean Teacher (**EMA over model parameters** instead of predictions)

Laine et al., ICLR 2017, Temporal Ensembling for Semi-Supervised Learning

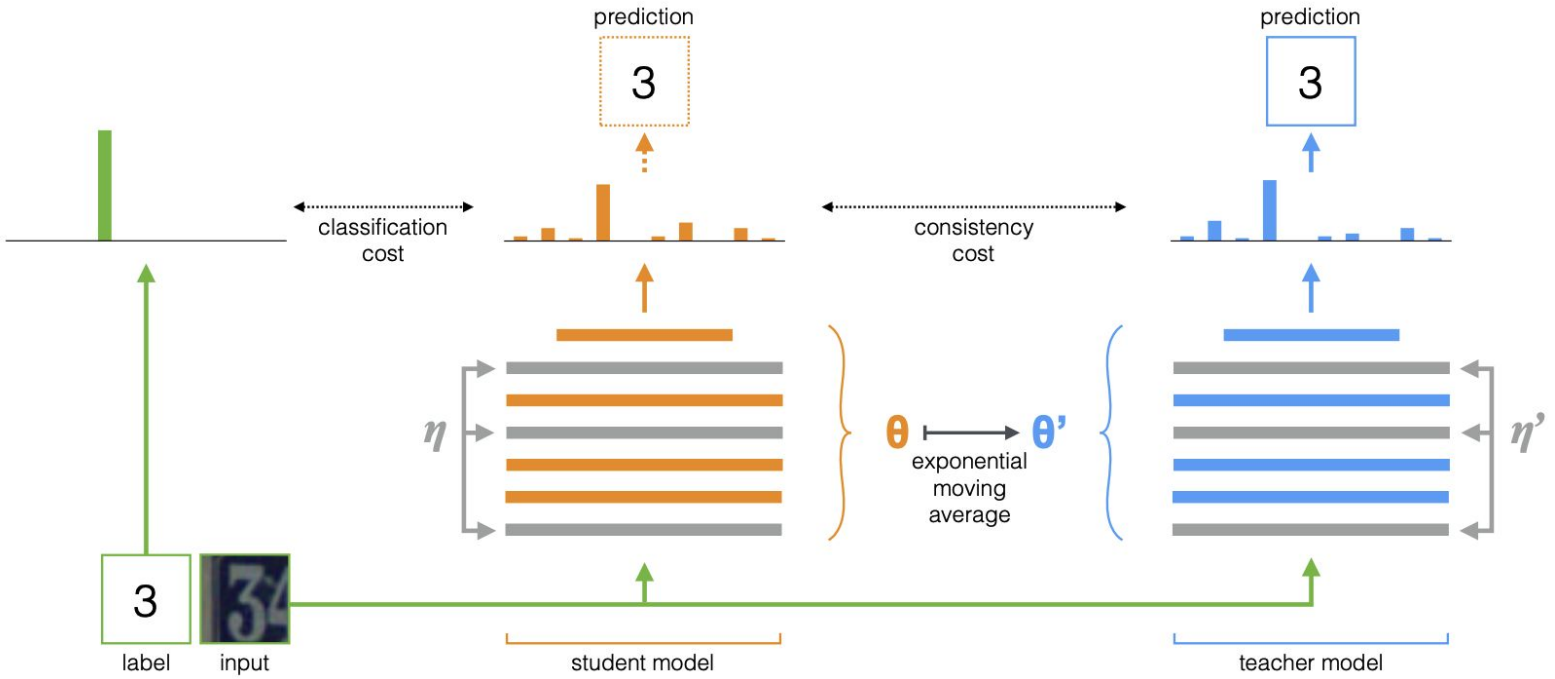
Mean Teachers

- Motivation
  - Temporal Ensembling stores an EMA prediction **for every training sample.**
    - Memory intensive
    - Targets are updated only **once per epoch**
- Core Idea
  - Replace the EMA of **predictions** with an EMA of **model weights.**
  - The teacher model is the exponential moving average of the student model:

$$\theta' \leftarrow \beta\theta' + (1 - \beta)\theta$$

Tarvainen et al., 2018, Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results

Mean Teachers



$$\theta' \leftarrow \beta\theta' + (1-\beta)\theta$$

Training Pipeline

- **Student**
  - Updated by gradient descent
  - Optimized with supervised and consistency losses
- **Teacher**
  - Receives the same input with different perturbations
  - Parameters updated only by EMA
  - No backpropagation

The consistency regularization loss is the distance between predictions by the student and teacher

Tarvainen et al., 2018, Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results

Mean Teachers

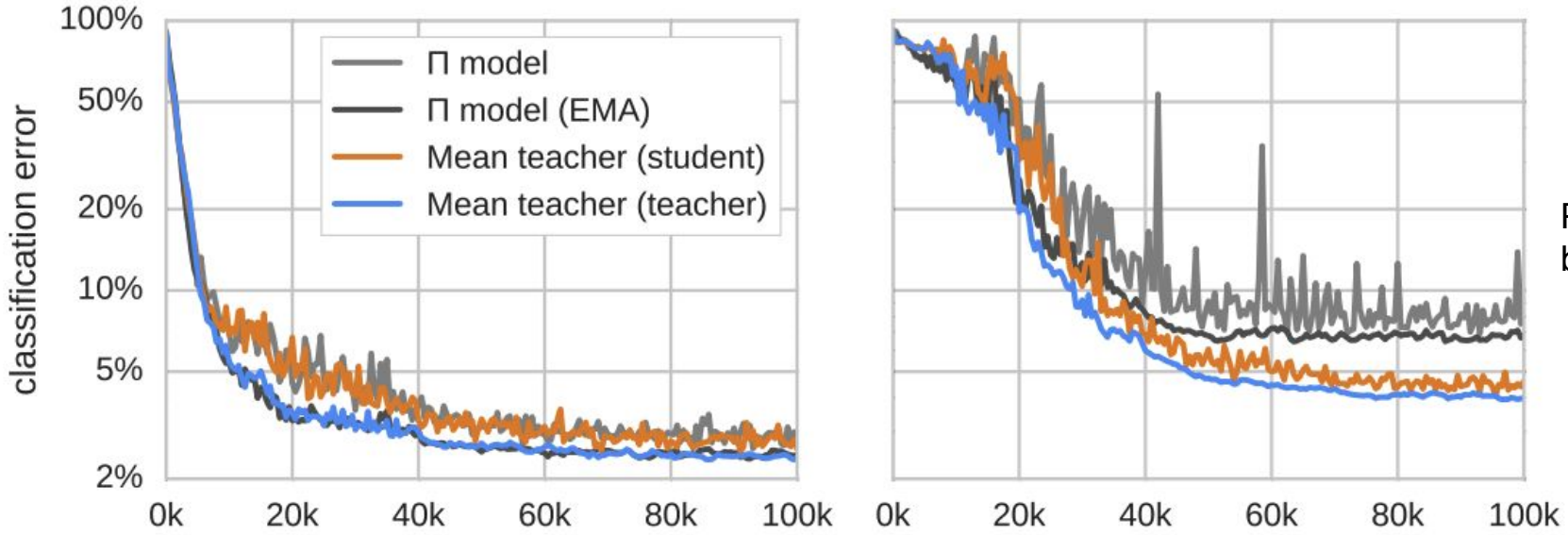
- Advantages
  - Stable teacher predictions
  - No per-sample prediction memory
  - Teacher updated every iteration
  - Scales well to large datasets

Learn from a stable teacher whose parameters are the EMA of the student.

Tarvainen et al., 2018, Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results

Mean Teachers

- Experiments
  - The mean teacher (in orange) has better performance than the student model.
  - The benefit is especially significant in the low-label regime.



Performance gap becomes much larger.

Classification error on SVHN of Mean Teacher and the Π Model.

Tarvainen et al., 2018, Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results

Mean Teachers

- Experiments

Table 1: Error rate percentage on SVHN over 10 runs (4 runs when using all labels). We use exponential moving average weights in the evaluation of all our models. All the methods use a similar 13-layer ConvNet architecture. See Table 5 in the Appendix for results without input augmentation.

	250 labels 73257 images	500 labels 73257 images	1000 labels 73257 images	73257 labels 73257 images
GAN [25]		18.44 ± 4.8	8.11 ± 1.3	
Π model [13]		6.65 ± 0.53	4.82 ± 0.17	2.54 ± 0.04
Temporal Ensembling [13]		5.12 ± 0.13	4.42 ± 0.16	2.74 ± 0.06
VAT+EntMin [16]			**3.86**	
Supervised-only	27.77 ± 3.18	16.88 ± 1.30	12.32 ± 0.95	2.75 ± 0.10
Π model	9.69 ± 0.92	6.83 ± 0.66	4.95 ± 0.26	2.50 ± 0.07
Mean Teacher	$\mathbf{4.35 \pm 0.50}$	$\mathbf{4.18 \pm 0.27}$	3.95 ± 0.19	$\mathbf{2.50 \pm 0.05}$

Tarvainen et al., 2018, Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results

Mean Teachers

- Effective Perturbations
  - Input augmentation is essential.
    - Random flip
    - Gaussian noise
  - Dropout is applied only to the **student** model.
  - Dropout is **not necessary** for the teacher model.
- EMA Decay (β)
  - Performance is sensitive to the EMA decay rate.
  - A practical strategy:
    - **Early** training: β=0.99
    - **Later** training: β=0.999
  - Increase the teacher stability as the student converges.
- Consistency Loss
  - **MSE** performs better than KL divergence.
  - MSE became the default consistency objective in many later methods.

Tarvainen et al., 2018, Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results

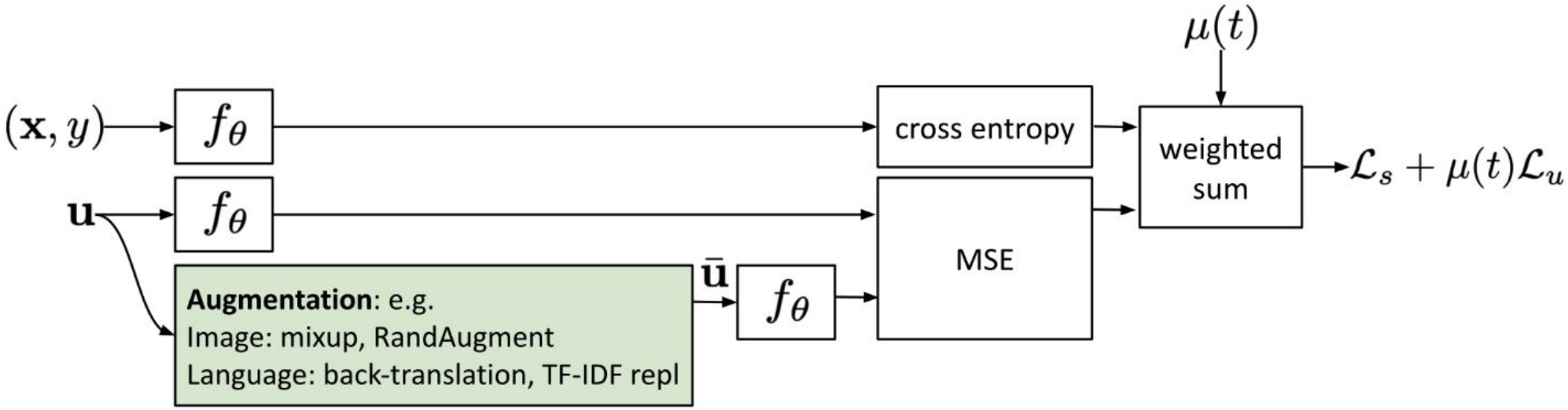
Mean Teachers

- Main Contributions
  - Replaced EMA of predictions with EMA of model parameters.
  - Provided more stable consistency targets.
  - Eliminated per-sample prediction storage.
  - Improved scalability to large datasets.

Tarvainen et al., 2018, Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results

Noisy Samples as Learning Targets

- Core Idea
  - Apply strong augmentations to an unlabeled sample.
  - The prediction should remain consistent before and after augmentation.



Key Insight

- The model should make the same prediction for an unlabeled sample and its augmented version.

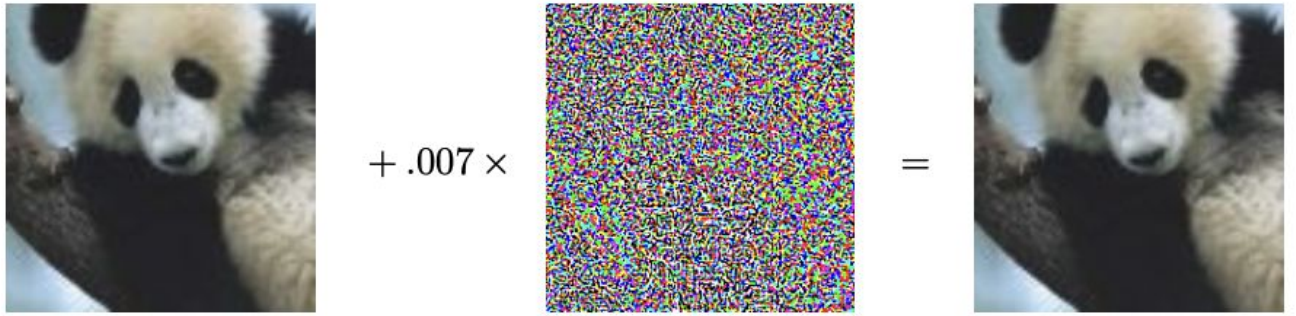
Noisy consistency training directly enforces prediction invariance between an input and its augmented version using the same model.

Noisy Samples as Learning Targets

- Compared with Π-model
  - Uses input augmentations on unlabeled data instead of stochastic perturbations inside the network.
- Compared with Mean Teacher
  - Uses self-consistency between original and augmented inputs instead of an EMA teacher as the learning target.

Noisy Samples as Learning Targets

- Adversarial Training (Goodfellow et al., 2014)
  - Add adversarial **perturbations** to the input during training.
  - Train the model to **produce the correct prediction** even under worst-case perturbations.
  - Improves robustness against **adversarial attacks**.



Goodfellow et al., ICLR 2015, Explaining and Harnessing Adversarial Examples

Noisy Samples as Learning Targets

- How it works
  - 1. Compute the adversarial perturbation that maximally changes the prediction.
  - 2. Add the perturbation to the input.
  - 3. Minimize the prediction difference between clean and perturbed inputs.
- Adversarial perturbation
  - Find the worst-case perturbation within a small norm constraint.
  - In practice, approximate it using the gradient direction (FGSM):

GT(label, one-hot) Model prediction for the attacked image

$$\mathcal{L}_{\text{adv}}(\mathbf{x}^l, \theta) = D[q(y \mid \mathbf{x}^l), p_\theta(y \mid \mathbf{x}^l + r_{\text{adv}})]$$

$$r_{\text{adv}} = \arg\max_{r;\|r\|\leq\epsilon} D[q(y \mid \mathbf{x}^l), p_\theta(y \mid \mathbf{x}^l + r_{\text{adv}})]$$

$$r_{\text{adv}} \approx \epsilon \frac{g}{\|g\|_2} \approx \epsilon \text{sign}(g) \quad \text{where } g = \nabla_r D[y, p_\theta(y \mid \mathbf{x}^l + r)]$$

Find the noise in the direction where the model is most confused.

Goodfellow et al., ICLR 2015, Explaining and Harnessing Adversarial Examples

Virtual Adversarial Training

- VAT extends the idea to work in semi-supervised learning.
  - The VAT loss applies to both labeled and unlabeled samples.

GT(label, one-hot)

$$\mathcal{L}_{\text{adv}}(\mathbf{x}^l, \theta) = D[q(y \mid \mathbf{x}^l), p_\theta(y \mid \mathbf{x}^l + r_{\text{adv}})]$$

If GT is known, ⇩ use current model prediction

$$\mathcal{L}_u^{\text{VAT}}(\mathbf{x}, \theta) = D[p_{\hat{\theta}}(y \mid \mathbf{x}), p_\theta(y \mid \mathbf{x} + r_{\text{vadv}})]$$

$$r_{\text{vadv}} = \arg\max_{r;\|r\| \le \epsilon} D[p_{\hat{\theta}}(y \mid \mathbf{x}), p_\theta(y \mid \mathbf{x} + r)]$$

Adversarial Training	VAT
Uses ground-truth labels	Uses model predictions
Supervised learning	Semi-supervised learning
Improves robustness	Enforces local smoothness of predictions

VAT replaces the ground-truth label with the model's own prediction, enabling adversarial consistency training on unlabeled data.

Miyato et al., TPAMI 2018, Virtual Adversarial Training: A Regularization Method for Supervised and Semi-Supervised Learning

Virtual Adversarial Training

TABLE 3: Test performance of semi-supervised learning methods on MNIST with the permutation invariant setting. The value N_l stands for the number of labeled examples in the training set. The top part cites the results provided by the original paper. The bottom part shows the performance achieved by our implementation. (PEA = Pseudo Ensembles Agreement, DGM = Deep Generative Models, FM=feature matching)

Method	Test error rate(%) $N_l = 100$	$N_l = 1000$
TSVM [8]	16.81	5.38
PEA [5]	5.21	2.87
DGM (M1+M2) [22]	3.33 (±0.14)	2.40 (±0.02)
CatGAN [37]	1.91 (±0.1)	1.73 (±0.18)
Skip DGM [27]	1.32 (±0.07)	
Ladder networks [33]	1.06 (±0.37)	**0.84** (±0.08)
Auxiliary DGM [27]	0.96 (±0.02)	
GAN with FM [36]	**0.93** (±0.07)	
RPT	6.81 (±1.30)	1.58 (±0.54)
VAT	1.36 (±0.03)	1.27 (±0.11)

TABLE 4: Test performance of semi-supervised learning methods on SVHN and CIFAR-10 *without* image data augmentation. The value N_l stands for the number of labeled examples in the training set. The top part cites the results provided by the original paper. The middle and bottom parts show the performance achieved by our implementation. The asterisk(*) stands for the results on the permutation invariant setting. (DGM=Deep Generative Models, FM=feature matching)

Method	Test error rate(%) SVHN $N_l = 1000$	CIFAR-10 $N_l = 4000$
SWWAE [48]	23.56	
*Skip DGM [27]	16.61 (±0.24)	
*Auxiliary DGM [27]	22.86	
Ladder networks, Γ model [33]		20.40 (±0.47)
CatGAN [37]		19.58 (±0.58)
GAN with FM [36]	8.11 (±1.3)	18.63 (±2.32)
∏ model [24]	5.43 (±0.25)	16.55 (±0.29)
(on Conv-Small used in [36])		
RPT	8.41 (±0.24)	18.56 (±0.29)
VAT	**6.83** (±0.24)	**14.87** (±0.13)
(on Conv-Large used in [24])		
VAT	5.77 (±0.32)	14.18 (±0.38)
VAT+EntMin	**4.28** (±0.10)	**13.15** (±0.21)

RPT smooths predictions in random directions, whereas VAT smooths only along the most adversarial direction, making the regularization significantly more effective.

Miyato et al., TPAMI 2018, Virtual Adversarial Training: A Regularization Method for Supervised and Semi-Supervised Learning

Interpolation Consistency Training

- ICT enhances the dataset
  - Following the idea of **MixUp**.
  - Add more interpolations of data points.
  - Expect the model prediction to be consistent with interpolations of the corresponding labels.
  - Uses a **Mean Teacher** to generate stable targets.

$$\text{mixup}_\lambda(\mathbf{x}_i, \mathbf{x}_j) = \lambda \mathbf{x}_i + (1-\lambda)\mathbf{x}_j$$ Two unlabeled samples are linearly interpolated

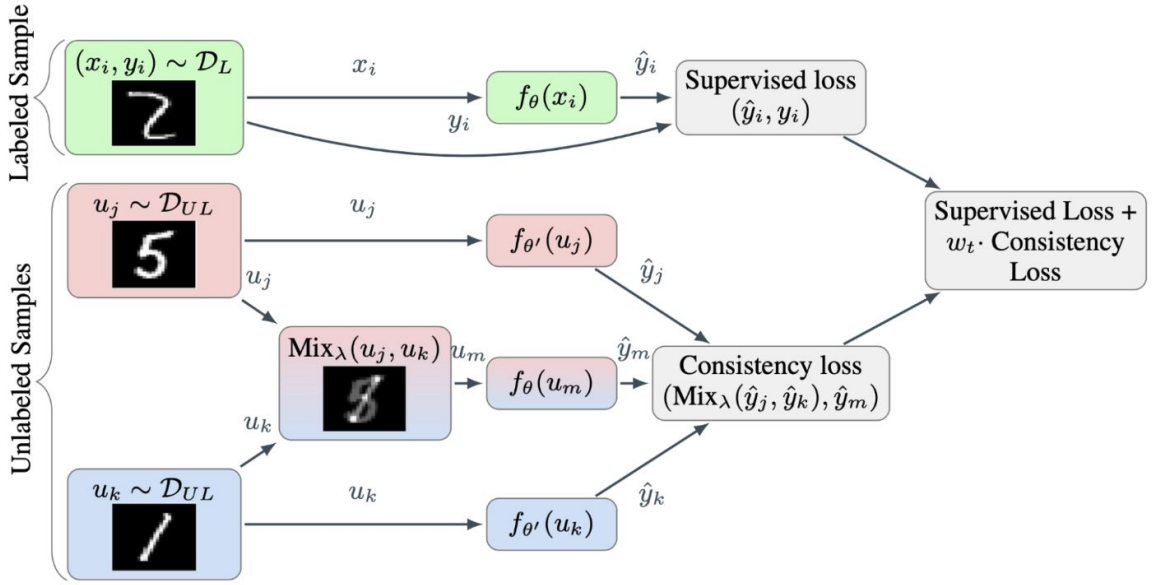
$$p(\text{mixup}_\lambda(y \mid \mathbf{x}_i, \mathbf{x}_j)) \approx \lambda p(y \mid \mathbf{x}_i) + (1-\lambda) p(y \mid \mathbf{x}_j)$$ Consistency Target

Mixing inputs should produce mixed predictions.

Verma et al., Neural Networks 2022, Interpolation Consistency Training for Semi-Supervised Learning

Interpolation Consistency Training

- ICT enhances the dataset
  - Random MixUp samples are likely to lie near the decision boundary.
  - Consistency on these interpolated samples encourages smoother decision boundaries.
  - Aligns well with the **Low-Density Separation Assumption.**



Mixing inputs should produce mixed predictions.

$$\mathcal{L}_u^{\text{ICT}} = \mathbb{E}_{\mathbf{u}_i, \mathbf{u}_j \sim \mathcal{U}} \mathbb{E}_{\lambda \sim \text{Beta}(\alpha,\alpha)} D[p_\theta(y \mid \text{mixup}_\lambda(\mathbf{u}_i, \mathbf{u}_j)), \text{mixup}_\lambda(p_{\theta'}(y \mid \mathbf{u}_i), p_{\theta'}(y \mid \mathbf{u}_j)]$$

Verma et al., Neural Networks 2022, Interpolation Consistency Training for Semi-Supervised Learning

Interpolation Consistency Training

Table 1
Error rates (%) on CIFAR-10 using CNN-13 architecture. We ran three trials for ICT.

Model	1000 labeled 50 000 unlabeled	2000 labeled 50 000 unlabeled	4000 labeled 50 000 unlabeled
Supervised	39.95 ± 0.75	31.16 ± 0.66	21.75 ± 0.46
Supervised (Mixup)	36.48 ± 0.15	26.24 ± 0.46	19.67 ± 0.16
Supervised (Manifold Mixup)	34.58 ± 0.37	25.12 ± 0.52	18.59 ± 0.18
Π model (Laine & Aila, 2017)	31.65 ± 1.20	17.57 ± 0.44	12.36 ± 0.31
TempEns (Laine & Aila, 2017)	23.31 ± 1.01	15.64 ± 0.39	12.16 ± 0.24
MT (Tarvainen & Valpola, 2017)	21.55 ± 1.48	15.73 ± 0.31	12.31 ± 0.28
VAT (Miyato et al., 2018)	–	–	11.36 ± NA
VAT+Ent (Miyato et al., 2018)	–	–	10.55 ± NA
VAdD (Park et al., 2018)	–	–	11.32 ± 0.11
SNTG (Luo et al., 2018)	18.41 ± 0.52	13.64 ± 0.32	10.93 ± 0.14
MT+ Fast SWA (Athiwaratkun et al., 2019)	15.58 ± NA	11.02 ± NA	9.05 ± NA
ICT	**15.48 ± 0.78**	**9.26 ± 0.09**	**7.29 ± 0.02**

Verma et al., Neural Networks 2022, Interpolation Consistency Training for Semi-Supervised Learning

Unsupervised Data Augmentation

- Unsupervised Data Augmentation (UDA)
  - Applies strong data augmentations to unlabeled samples.
  - Encourages consistent predictions between the original and augmented images.
  - Emphasizes the importance of **augmentation quality** rather than adversarial perturbations.

$$p(y|x) \approx p(y|\mathrm{Aug}(x))$$

Xie et al., NeurIPS 2020, Unsupervised Data Augmentation for Consistency Training

Unsupervised Data Augmentation

- Unsupervised Data Augmentation (UDA)
  - What makes a good augmentation?
  - A good augmentation should be
    - Label-preserving
    - Diverse
    - Realistic
    - Task-relevant

Crucial to use advanced data augmentation methods

Xie et al., NeurIPS 2020, Unsupervised Data Augmentation for Consistency Training

Unsupervised Data Augmentation

- Key Observation
  - Simply replacing adversarial perturbations with **strong semantic augmentations** substantially improves performance.
  - UDA outperforms previous consistency-based methods, including **VAT** and **ICT**.
  - Demonstrates that **augmentation quality is a key factor** in semi-supervised learning.

Method	Model	# Param	CIFAR-10 (4k)	SVHN (1k)
Π-Model [32]	Conv-Large	3.1M	12.36 ± 0.31	4.82 ± 0.17
Mean Teacher [58]	Conv-Large	3.1M	12.31 ± 0.28	3.95 ± 0.19
VAT + EntMin [41]	Conv-Large	3.1M	10.55 ± 0.05	3.86 ± 0.11
SNTG [35]	Conv-Large	3.1M	10.93 ± 0.14	3.86 ± 0.27
ICT [60]	Conv-Large	3.1M	7.29 ± 0.02	3.89 ± 0.04
Pseudo-Label [33]	WRN-28-2	1.5M	16.21 ± 0.11	7.62 ± 0.29
LGA + VAT [25]	WRN-28-2	1.5M	12.06 ± 0.19	6.58 ± 0.36
ICT [60]	WRN-28-2	1.5M	7.66 ± 0.17	3.53 ± 0.07
MixMatch [3]	WRN-28-2	1.5M	6.24 ± 0.06	2.89 ± 0.06
Mean Teacher [58]	Shake-Shake	26M	6.28 ± 0.15	-
Fast-SWA [1]	Shake-Shake	26M	5.0	-
MixMatch [3]	WRN	26M	4.95 ± 0.08	-
UDA (RandAugment)	WRN-28-2	1.5M	4.32 ± 0.08	**2.23 ± 0.07**
UDA (RandAugment)	Shake-Shake	26M	3.7	-
UDA (RandAugment)	PyramidNet	26M	**2.7**	-

RandAugment

- Rotate
- Translate
- Shear
- Color
- Brightness
- Contrast
- Posterize
- Solarize
- Equalize
- ...

Xie et al., NeurIPS 2020, Unsupervised Data Augmentation for Consistency Training

Unsupervised Data Augmentation

- Why is UDA Better?
  - Uses **strong, label-preserving augmentations** (RandAugment).
  - Produces more diverse training samples.
  - Encourages consistency under realistic semantic variations instead of small adversarial perturbations.

How we perturb the input matters more than simply adding perturbations.
High-quality semantic augmentations lead to much stronger consistency regularization.

Xie et al., NeurIPS 2020, Unsupervised Data Augmentation for Consistency Training

Can a more accurate teacher support stronger perturbations?

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Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

Perturbed and Strict Mean Teachers for Semi-supervised Semantic Segmentation

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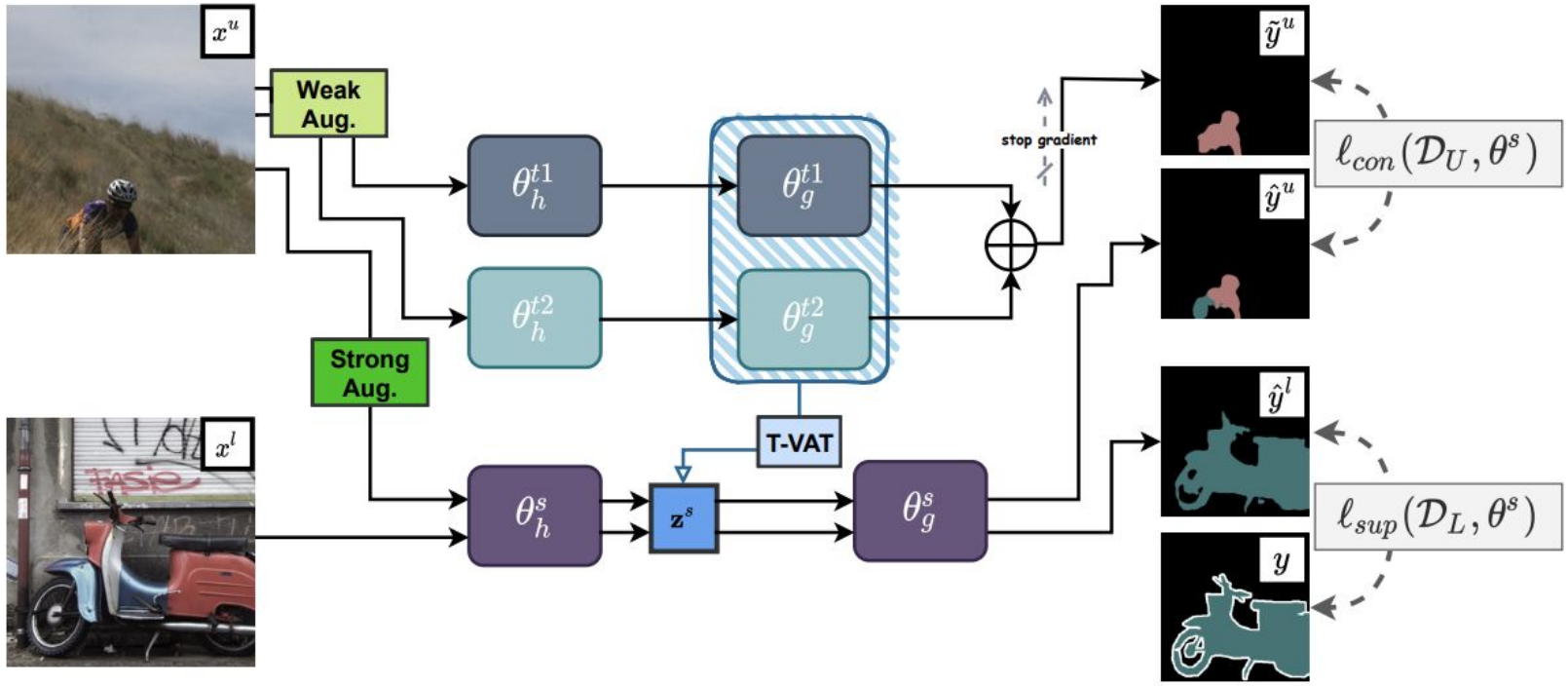
Liu et al., CVPR 2022, Perturbed and Strict Mean Teachers for Semi-supervised Semantic Segmentation

Strong Perturbations Need Reliable Targets

- Mean Teacher uses teacher predictions as training targets for unlabeled images.
- **Strong perturbations** provide more challenging examples and can improve **generalization**.
- However, teacher predictions are **not always correct**.
- If the **teacher** is **wrong**, the student learns to reproduce the **same error** under strong perturbations.
- This can **reinforce incorrect predictions** during training.

Before applying stronger perturbations, we need more reliable teacher predictions.

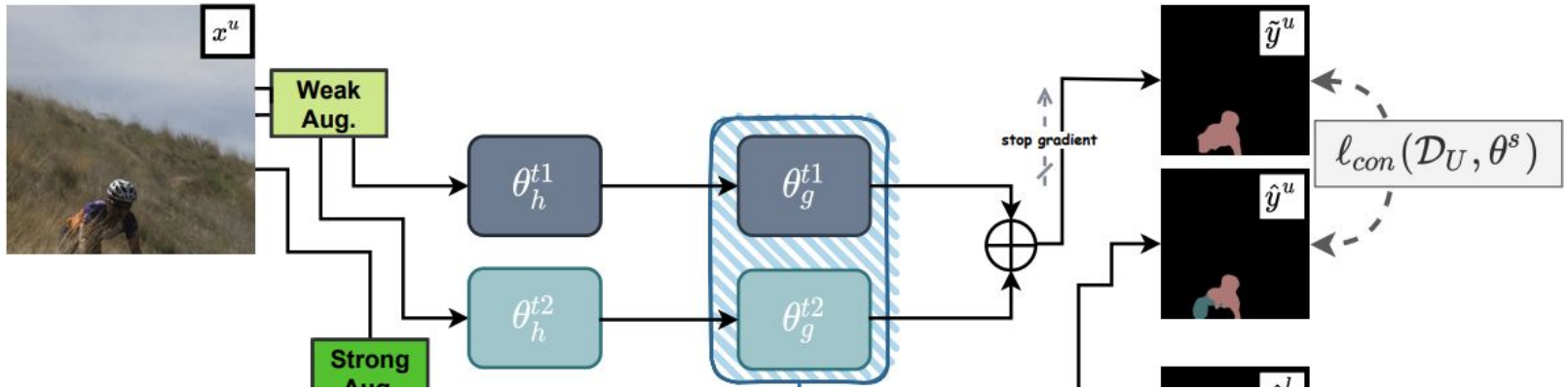
Perturbed and Strict Mean Teachers



- Labeled data
  - Train the student with ground-truth masks.
- Unlabeled data
  - Two teachers generate the target.
  - The student receives stronger perturbations.
  - The student matches the teacher ensemble.

1. Two Mean Teachers
2. Teacher Ensemble
3. Strongly Perturbed Student
4. Strict Consistency Loss

A More Reliable Teacher



$$\hat{\mathbf{y}} = \sigma(0.5 \times (f_{\theta^{t1}}(\mathbf{x}) + f_{\theta^{t2}}(\mathbf{x})))$$

$$\theta^k = \gamma \times \theta^k + (1 - \gamma) \times \theta^s$$

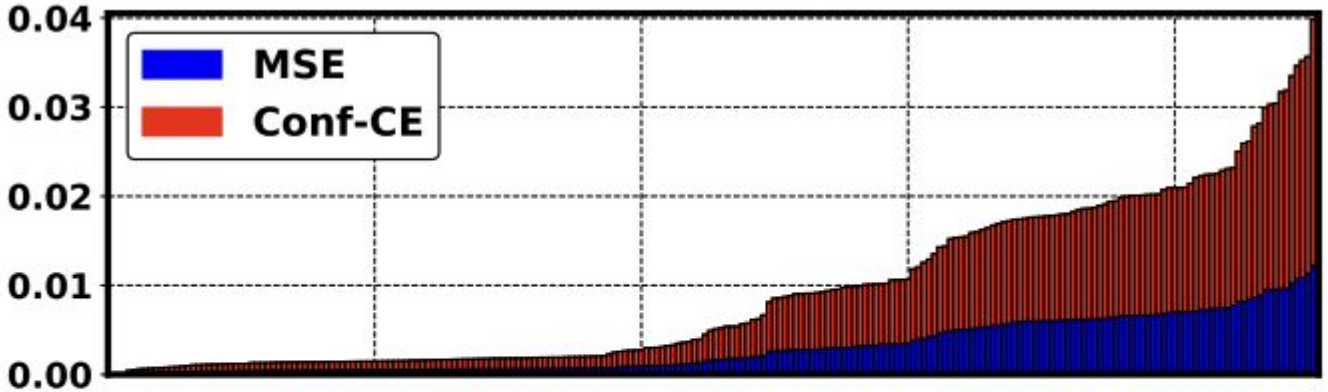
- Only one teacher is updated in each epoch.
- The two teachers maintain different temporal histories.
- Their ensemble reduces prediction variance.

Each teacher averages the student over time. The method then averages the two teachers.

From MSE to Strict Consistency

- MSE loss
  - Robust to noisy teacher predictions
  - Small gradients near convergence
  - Relatively weak supervision
- Conf-CE

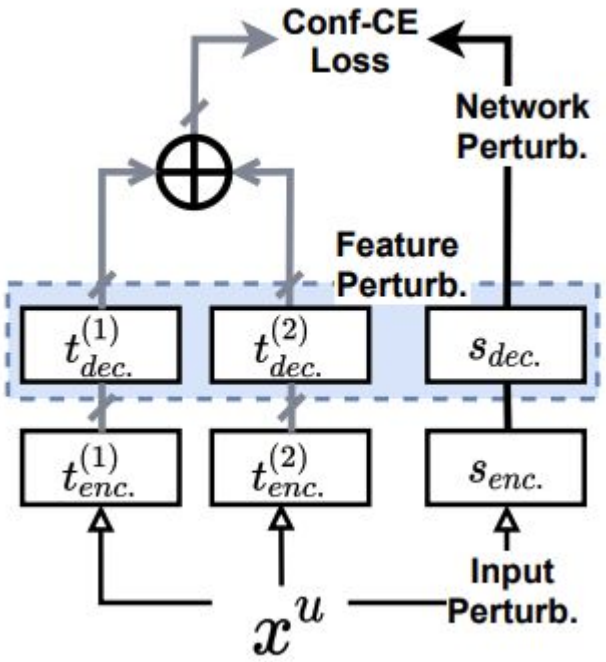
$$\ell_{con}(\mathcal{D}_U, \theta^s) = \frac{1}{|\mathcal{D}_U||\Omega|} \sum_{\mathbf{x}\in\mathcal{D}_U} \sum_{\omega\in\Omega} c(\omega)\ell(\tilde{\mathbf{y}}(\omega), p_{\theta^s}(\mathbf{x})(\omega)),$$



More accurate teachers allow stronger cross-entropy supervision.

Three Levels of Perturbation

- 1. Input Perturbation
  - Weak augmentation for teachers
  - Strong augmentation for the student
  - CutMix and Zoom In/Out
- 2. Feature Perturbation
  - Teacher-guided adversarial perturbation
  - Applied to the student feature
- 3. Network Perturbation
  - Two temporal teachers
  - One gradient-updated student
  - Different model predictions



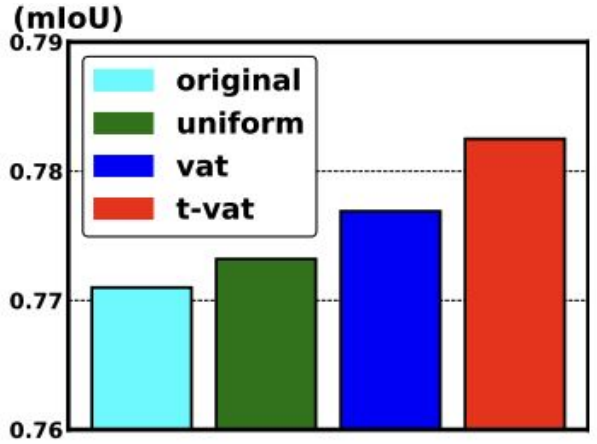
Consistency is enforced across input, feature, and network variations.

Teacher-Guided VAT

- Standard VAT.
  - The student finds its own adversarial direction.
  - Early student predictions can be inaccurate.
- T-VAT
  - The student generates a feature z_s.
  - Two teachers evaluate the feature.
  - Search for the direction that maximizes changes to the teacher predictions.
  - Apply the corresponding perturbation to the student feature.
  - Train the student to maintain the teacher targets.

$$\text{maximise} \; d\Big(\sigma\big(0.5 \times (g_{\theta_g^{t1}}(\mathbf{z}^s) + g_{\theta_g^{t2}}(\mathbf{z}^s))\big), \; \sigma\big(0.5 \times (g_{\theta_g^{t1}}(\mathbf{z}^s + \mathbf{r}_{adv}) + g_{\theta_g^{t2}}(\mathbf{z}^s + \mathbf{r}_{adv}))\big)\Big),$$

$$\text{subject to} \; ||\mathbf{r}_{adv}||_2 <= \epsilon,$$



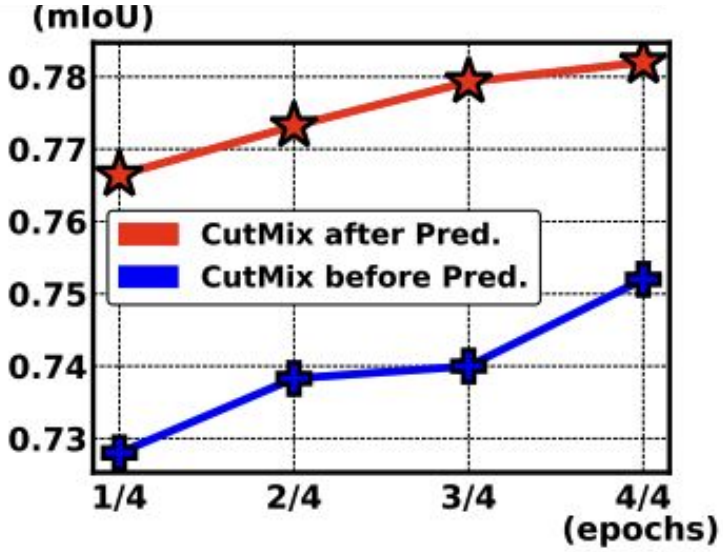
Teachers find the perturbation. The student learns from it.

CutMix After Prediction

- CutMix before prediction
  - Artificial boundaries appear in the mixed image.
  - Teacher predictions can contain mixing artifacts.
- CutMix after prediction
  - First, predict each image, then blend the results.
  - Teacher sees natural images

$$\ell(\mu(\tilde{\mathbf{y}}_i, \tilde{\mathbf{y}}_j, \mathbf{m}), p_{\theta^s}(\mu(\mathbf{x}_i, \mathbf{x}_j, \mathbf{m})))$$

$$\mu(\mathbf{x}_i, \mathbf{x}_j, \mathbf{m}) = (1 - \mathbf{m}) \odot \mathbf{x}_i + \mathbf{m} \odot \mathbf{x}_j$$



Mixing clean predictions gives a cleaner target than predicting a mixed image.

What Improves the Performance?

- Conf-CE provides the largest single gain.
- T-VAT adds effective feature perturbation.
- The auxiliary teacher further improves the target.
- The components provide complementary improvements.

MT	conf-CE	T-VAT	AT	Backbone	
				ResNet-50	ResNet-101
✓				71.49	73.50
✓	✓			73.79	76.39
✓	✓	✓		74.87	77.36
✓	✓	✓	✓	**75.70**	**78.20**

Better targets make stronger consistency learning possible.

Results and Takeaways

- The improvement is consistent across label ratios.
- The gain is larger when fewer labels are available.

Table 1. **Comparison with SOTA on Pascal VOC 2012.** All approaches are based on the DeeplabV3+ architecture, under different partition protocols [9]. The $*$ indicates the approaches re-implemented by [9]. Best results are in bold.

Method	ResNet-50				ResNet-101			
	1/16(662)	1/8(1323)	1/4(2646)	1/2(5291)	1/16(662)	1/8(1323)	1/4(2646)	1/2(5291)
MT* [37]	66.77	70.78	73.22	75.41	70.59	73.20	76.62	77.61
French et al.* [14]	68.90	70.70	72.46	74.49	72.56	72.69	74.25	75.89
CCT* [34]	65.22	70.87	73.43	74.75	67.94	73.00	76.17	77.56
GCT* [21]	64.05	70.47	73.45	75.20	69.77	73.30	75.25	77.14
ECS [31]	-	67.38	70.70	72.89	-	-	-	-
CPS [9]	71.98	73.67	74.90	76.15	74.48	76.44	77.68	78.64
CAC [22]	70.10	72.40	74.00	-	72.40	74.60	76.30	-
Ours	**72.83**	**75.70**	**76.43**	**77.88**	**75.50**	**78.20**	**78.72**	**79.76**

Results and Takeaways

- The teacher ensemble produces better boundaries.

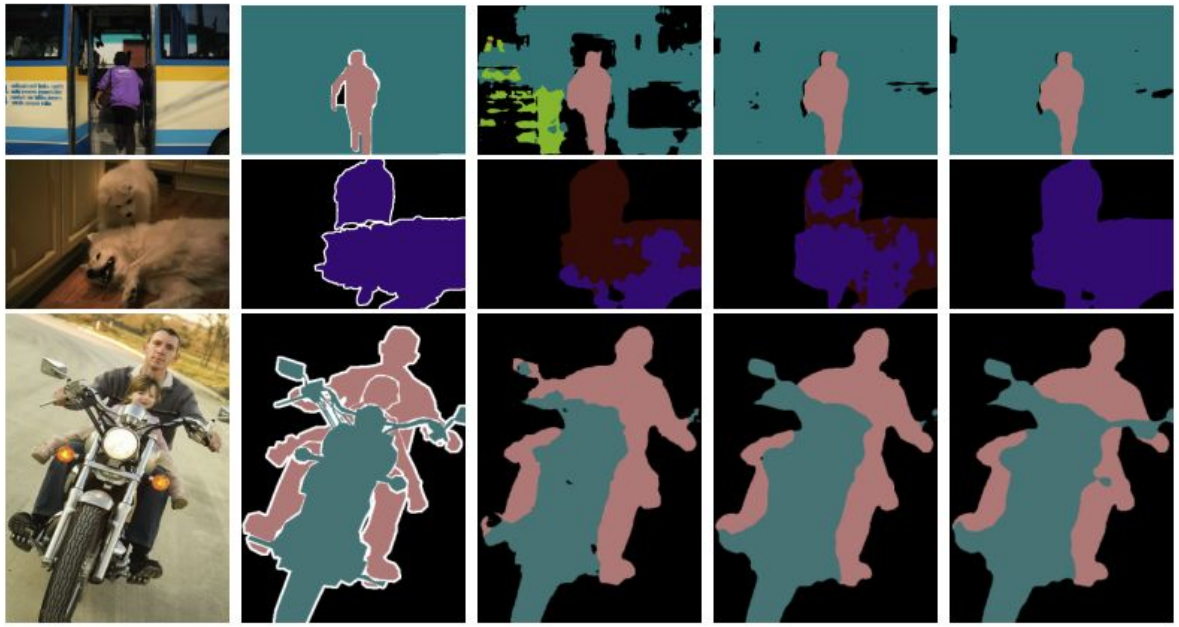


Figure 6. **Qualitative results** from *Pascal VOC 2012*. (a) input images, (b) ground truth, (c) supervised baseline results, (d) student results and (e) results by our approach.

What if the teacher becomes too similar to the student?

Switching Temporary Teachers for Semi-Supervised Semantic Segmentation

Jaemin Na[1], **Jung-Woo Ha**[2], **Hyung Jin Chang**[3], **Dongyoon Han**[2†], **Wonjun Hwang**[1,2†]
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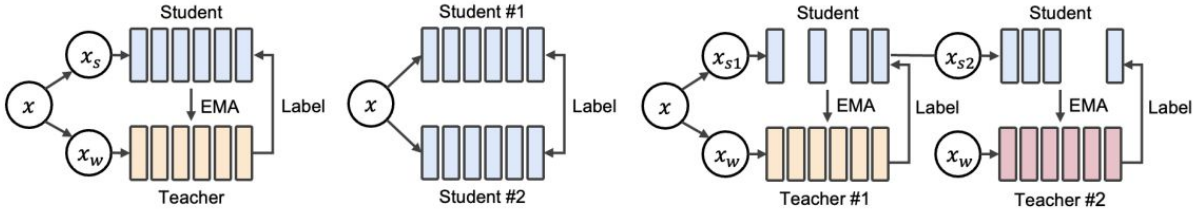
Na et al., NeurIPS 2023, Switching Temporary Teachers for Semi-Supervised Semantic Segmentation

What if the teacher becomes too similar to the student?

- Mean Teacher updates the teacher with an exponential moving average of the student.
- The teacher gradually follows the student throughout training.
- Their parameters and predictions can become increasingly similar.
- A highly similar teacher provides little new information.
- The teacher may also inherit the student’s errors.

Teacher–Student Coupling

- Single Teacher
  - One student and one permanent EMA teacher
  - The teacher continuously follows the same student
- Dual Student
  - Two independently trained students
  - Higher training and memory cost
- Dual Teacher
  - One student and two temporary teachers
  - Only one teacher guides the student at a time

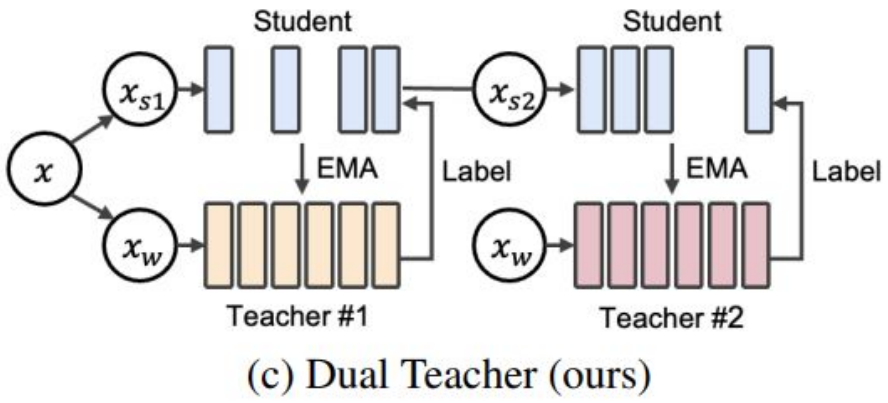


(a) Teacher-Student (b) Dual Student (c) Dual Teacher (ours)

Dual Teacher introduces diversity without training a second student.

Switching Temporary Teachers

- Two EMA teachers take turns guiding one student.
- Only the active teacher generates the target for the current epoch.
- The active teacher is updated from the current student.
- The student receives a different strong augmentation when the teacher changes.



(c) Dual Teacher (ours)

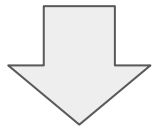
Epoch	Active teacher	Student augmentation
1	Teacher 1	ClassMix
2	Teacher 2	CutMix
3	Teacher 1	ClassMix or CutMix
4	Teacher 2	The other augmentation

Each teacher preserves the student's characteristics from different training stages.

What Changes from Mean Teacher?

- EMA of Dual Teacher

$$\theta_t \leftarrow \alpha\theta_t + (1-\alpha)\theta_s$$



$$k_e \in \{1, 2\}$$

$$\theta_{t_{k_e}} \leftarrow \alpha\theta_{t_{k_e}} + (1-\alpha)\theta_s$$

The EMA rule remains the same. The teacher schedule changes.

Training the Student

- The active teacher receives a weakly augmented image.
- It generates a pixel-level target.
- The student receives a strongly augmented version.
- The student learns to match the teacher's target.

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \lambda_u \mathcal{L}_{\text{unsup}}$$

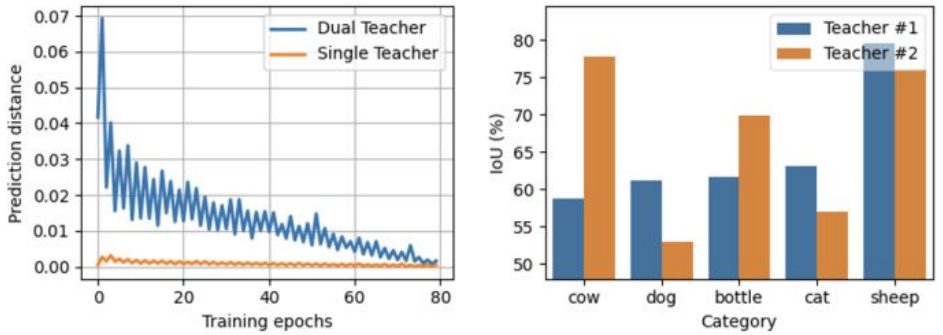
$$\mathcal{L}_{\text{unsup}} = \frac{1}{B_u HW} \sum_{i=1}^{B_u} \sum_{j=1}^{HW} \mathcal{L}_{\text{CE}} \left(p_{ij}^u, \hat{y}_{ij}^u \right)$$

- B_u: number of unlabeled images
- $H \times W$: number of pixels per image
- p_{ij}^u: student prediction for pixel j
- $\hat{y}_{ij}^u$: target from the active teacher
- λ_u: weight of the unlabeled loss

The objective resembles Mean Teacher. The supervision source changes over time.

Does Switching Actually Prevent Coupling?

- Figure 2(a): Prediction distance
  - Single Teacher stays very close to the student.
  - Dual Teacher maintains a larger prediction distance.
  - Switching prevents one teacher from continuously following the student.
- Figure 2(b): Class-wise IoU
  - The two teachers have similar overall mIoU.
  - However, their strengths differ across classes.
  - They can provide complementary supervision.



(a) Prediction distance (b) Class-wise IoU

Teacher diversity means different predictions, not simply different parameter values.

Why Switch Instead of Ensemble?

- Teacher ensemble
  - Both teachers guide the student simultaneously.
  - Their predictions are combined every epoch.
  - They can gradually lose their distinct characteristics.
- Teacher switching
  - One teacher guides the student at each epoch.
  - Each teacher observes a different student state.
  - Their temporal roles remain separated.

Table 7: Performance analysis of different teaching strategies on **PASCAL VOC 2012** across various partitions. Labeled images are sampled from the *blender* set and all methods are based on DeepLabv3+ with ResNet-50.

Method	1/16 (186)	1/8 (372)	1/4 (744)	1/2 (1,488)
Switching teachers	77.00	79.23	78.94	80.18
Ensembling teachers	73.25	75.99	76.41	77.34

Two teachers help only when they preserve different perspectives.

How Many Teachers Are Needed?

- Two teachers outperform one teacher.
- A third teacher provides no additional gain.
- Two augmentation types perform better than one.
- More augmentation types do not guarantee better performance.

Table 5: The performance relative to the number of teacher models and augmentations on $^1/_{16}$ splits of the PASCAL VOC *blender* set.

Method	Single	Dual	Triple
$Aug \times 1$	73.63	75.34	75.30
$Aug \times 2$	74.21	**77.00**	76.56
$Aug \times 3$	74.49	76.25	75.94

The best result comes from two teachers and two augmentation types.

Takeaway: Preserve a Useful Gap

- Mean Teacher
  - Stabilizes targets through temporal averaging
  - May gradually couple the teacher and student
- Dual Teacher
  - Alternates two temporary EMA teachers
  - Preserves different temporal perspectives
  - Provides complementary supervision to one student

A teacher should be stable, but it should not become identical to the student.

Are image-level perturbations enough for a Vision Transformer?

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Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

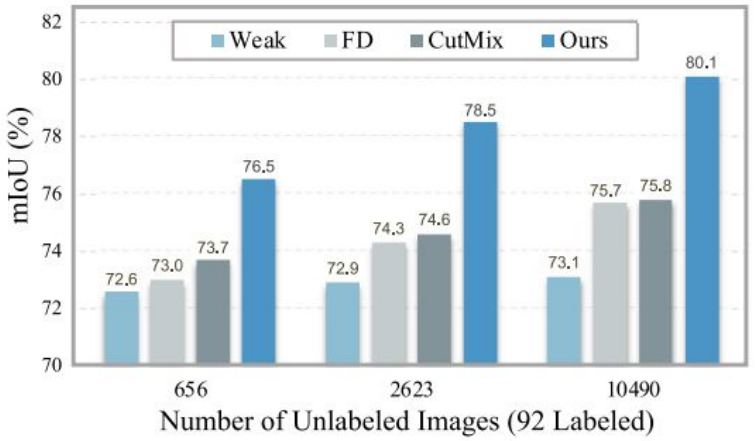
Training Vision Transformers for Semi-Supervised Semantic Segmentation

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Hu et al., CVPR 2024, Training Vision Transformers for Semi-Supervised Semantic Segmentation.

Are Image-Level Perturbations Enough for a Vision Transformer?

- Most consistency methods perturb the input image.
- The teacher receives a weakly augmented image.
- The student receives a strongly augmented image.
- However, a Vision Transformer processes images as patch tokens.
- Its self-attention and output distribution provide additional places for regularization.

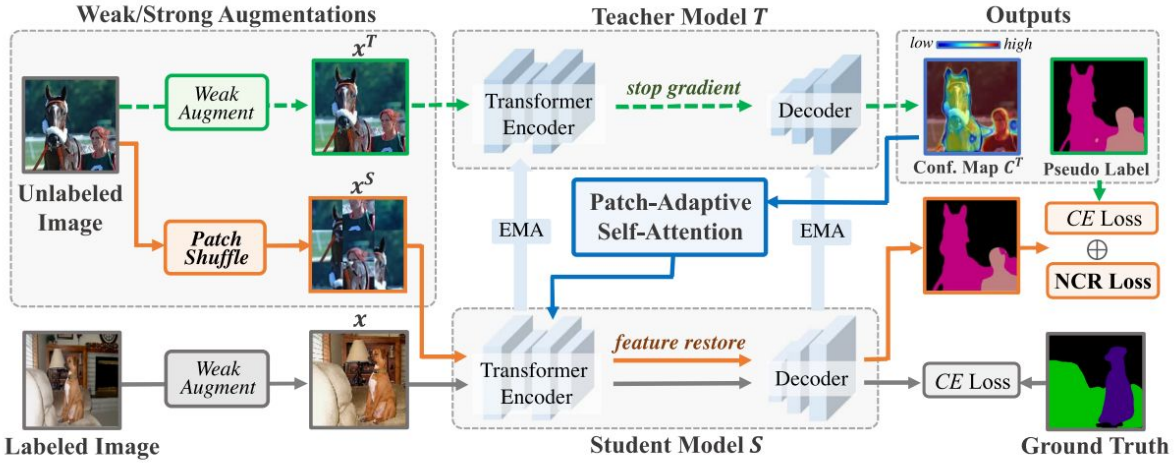


FD: feature dropout

Can we regularize the input, feature, and output spaces together?

S⁴Former: Three Levels of Regularization

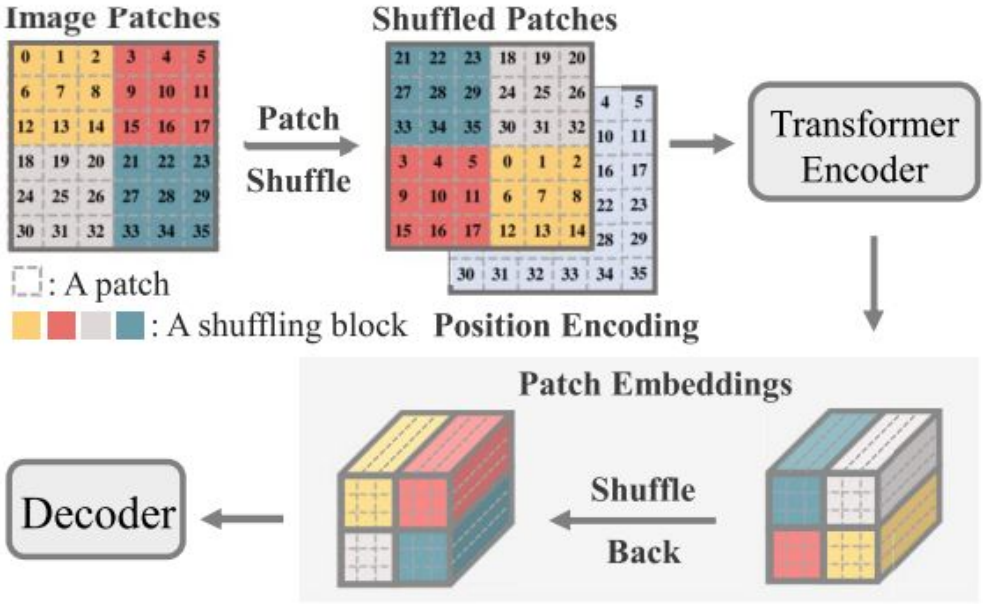
- Teacher
  - Receives a weakly augmented image
  - Generates pixel-level targets
  - Updated by EMA
- Student
  - Receives a strongly augmented image
  - Uses PatchShuffle and PASA
  - Matches both positive and negative class information



S⁴Former regularizes the Transformer at the input, feature, and output levels.

Input Regularization: PatchShuffle

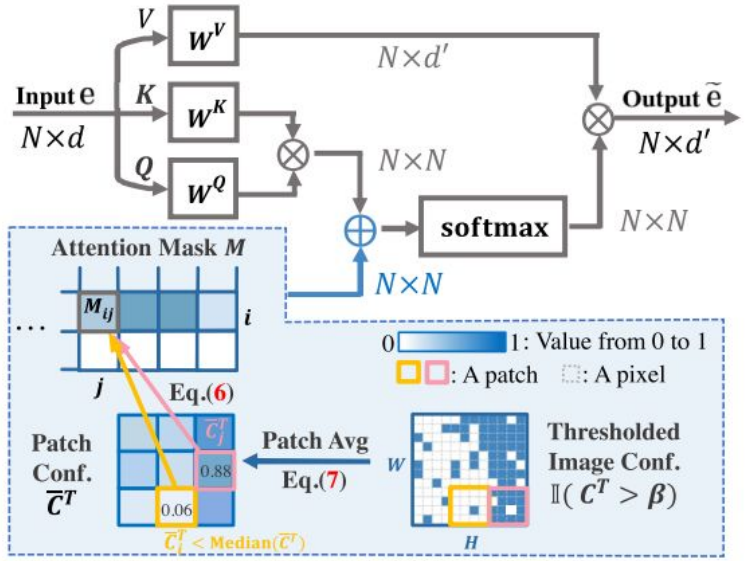
- Divide the image into an L x L patch grid.
- Group neighboring patches into M x M blocks.
- Shuffle the blocks before Transformer encoding.
- Restore the feature order before decoding.



The student must preserve the same semantics even when the global patch layout changes.

Feature Regularization: PASA

- The teacher estimates confidence for each image patch.
- High-confidence patches keep their original attention.
- Low-confidence patches receive additional attention modulation.
- Uncertain regions interact more strongly with other uncertain regions.

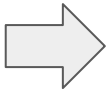


PASA perturbs attention where the teacher is uncertain.

Output Regularization: Negative Class Ranking

- CE matches the most likely class.
- NCR matches the relative distribution over the remaining classes.
- Similar classes receive similar relative probabilities.
- The student learns more than a hard pseudo-label.
- Recalculate the softmax excluding the pseudo-label class.

$$\mathcal{L}_{CE} = -\log\left(\frac{\exp(p^S_{\tilde{y}})}{\sum_{k=1}^{K}\exp(p^S_k)}\right)$$



$$\mathcal{L}_{NCR} = \sum_{k\neq\tilde{y},k\in K}\left(\hat{p}^T_k - \hat{p}^S_k\right)^2, \quad \text{where}$$

$$\hat{p}^{T\text{ or }S}_k = \frac{\exp(p^{T\text{ or }S}_k)}{\sum_{k\neq\tilde{y},k\in K}\exp(p^{T\text{ or }S}_k)}$$

The teacher provides a class relationship, not only a single target class.

Do All Three Components Help?

- Each component improves the Transformer baseline.
- PatchShuffle provides the largest individual improvement.
- PASA and NCR provide complementary gains.
- Combining all components gives the best result.

Methods	Split = 1/16	1/8	1/4
SupOnly	67.7	72.8	77.4
Base	75.8	77.4	80.0
+PatchShuffle	78.4 (+2.6)	80.2 (+2.8)	82.1 (+2.1)
+PASA	77.9 (+2.1)	79.4 (+2.0)	81.3 (+1.3)
+NCR	77.5 (+1.7)	78.7 (+1.3)	81.9 (+1.9)
+All	80.1 (+4.3)	81.3 (+3.9)	82.4 (+2.4)

The largest gain is observed when labeled data are most limited.

What Does S⁴Former Add to Mean Teacher?

- What remains unchanged
  - Teacher–student framework
  - EMA teacher update
  - Weak-to-strong consistency
- What changes
  - Patch-aware input perturbation
  - Confidence-aware attention perturbation
  - Class-distribution regularization

Level	Standard Mean Teacher	S⁴Former
Input	Weak and strong image augmentation	PatchShuffle
Feature	Standard feature extraction	PASA
Output	Positive pseudo-label consistency	Negative class ranking
Teacher update	EMA	EMA

Effective consistency regularization should reflect the structure of the model.

Summary

- PS-MT: How can the teacher target be made more reliable?
- Dual Teacher: How can we prevent the teacher from becoming too similar to the student?
- S⁴Former: To which part of the Transformer should consistency be applied?

Next Week..

- Pseudo Labeling ..